

Final Project

Reviewing CCUS (Carbon Capture, Utilisation and Storage) as a technological alternative in decarbonising Indian industries, from an environmental, economic and policy perspective.



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Abstract

The world is experiencing climate change and carbon emissions are being identified as the major cause for this. Industrial emissions form a large part of these emissions and the technologies to abate these emissions are in an infant stage. Carbon Capture, Storage, and Utilization has been recognized as a technological alternative that can solve the problem at a scale in the future. In India, these technologies are almost non-existent while the policy continues to be ambitious on decarbonising its industries. In this project, we introduce CCUS as a technology to decarbonise industries and assess its applicability in India from environmental, social, economic, and policy perspectives. We observe that the data and research is unavailable due to lack of co-ordinated research in different aspects of the technology.

Objectives

- To introduce the problem of rising industrial emissions and the distribution of these emissions
- Reviewing CCUS as a technological solution to decarbonise Industries
- Identifying the environmental, social, and economic risks attached to CCUS, especially in the Indian context.

Rising Industrial Emissions – The Problem

Rising Emissions Globally

The advent of industrialization brought about an unprecedented increase in economic growth; what it also brought along with it, vastly neglected, was an increase in harmful pollutants into the atmosphere. These pollutants are not limited to just one country or a group of countries, but part of a global phenomena. Lost in the race of the “quicker we set up our industries, the better off our people will be”, slow irreversible and invisible changes started to creep on us. Figure 1 is a snippet from the IPCC 6th assessment report – We can see that emissions from burning of fossil fuels and land use change pattern have been increasing rapidly, subsequently this results in a higher concentration of greenhouse gasses in the atmosphere; this intensifies the greenhouse effect as more and more of the sun’s heat is trapped – resulting in an increasing in temperature. Over the course of millions of years, temperatures have fluctuated and so have carbon emissions however, these changes had taken place over millions of years – making such sharp increases in temperatures and carbon emissions much of a concern,

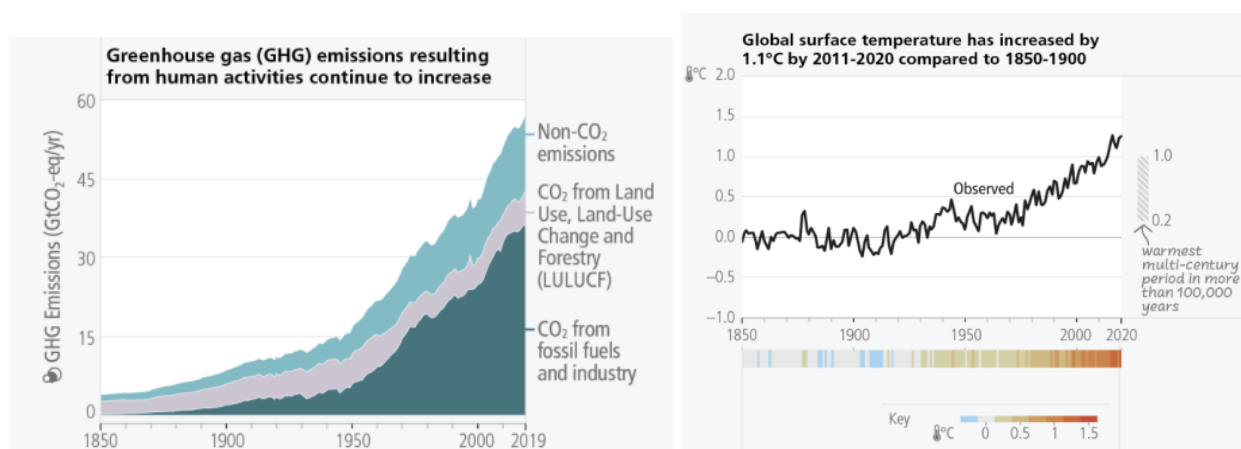


Figure 1: Greenhouse gas emissions from human activities have been increasing.

Source: IPCC 6th Assessment Report

Furthermore, carbon emissions can be broken down into two categories – natural and anthropogenic. Natural carbon emissions are those emissions which occur naturally, i.e, through volcanic eruptions, respiration or decomposition. These natural emissions emit carbon dioxide, which is a greenhouse gas; however, these natural emissions are usually balanced out through nature itself in the form of carbon sinks such as forests. On the other hand, anthropogenic emissions are those emissions which are released post the interaction of humans with the natural

resources – Burning of fossil fuels, deforestation, industrialization are some examples. These activities release large amounts of carbon dioxide into the atmosphere – contributing to climate change greatly. In addition, the past trends and future projections of carbon emissions raise major concerns – carbon emissions arising from both industrial activities and burning of fossil fuels have been increasing at an increasing rate and are projected to do so as well. Figure 2 and 3¹ indicate that since the early 1900s emissions have been growing, and growing at an increasing rate; it is also projected that they will continue to rise at an increasing rate in the future.

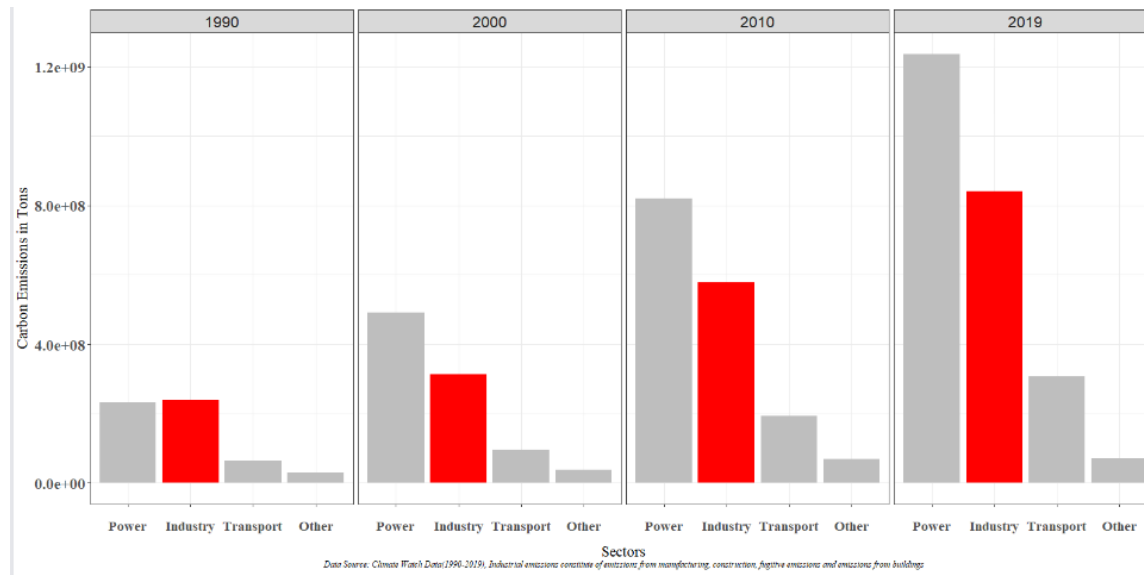


Figure 2: An Increasing Trend in industrial emissions in India from 1990 to the 2020s
Data Source: Climate Watch Data

Increase in carbon emissions will have a detrimental impact on society and biodiversity. The living planet report published by WWF reports an average decline of 69% in species populations since 1970. The climate change crisis is a “wicked problem” – its effects are unevenly distributed across the population; the vulnerable societies and people – such as farmers – are going to experience greater detrimental effects on their livelihood. Hence, technological solutions – CCUS, Renewable Energy etc – to limit and reduce our carbon emissions play a pivotal role in deciding the future of our society. In the subsequent section of this report, we will talk about the proposed solution as well as focus, in particular on one technological solution – CCUS (Carbon Capture Utilization and Storage).

¹ Figure 3 contains the sectoral composition of industrial emissions as well, This will be discussed in later sections of the report.

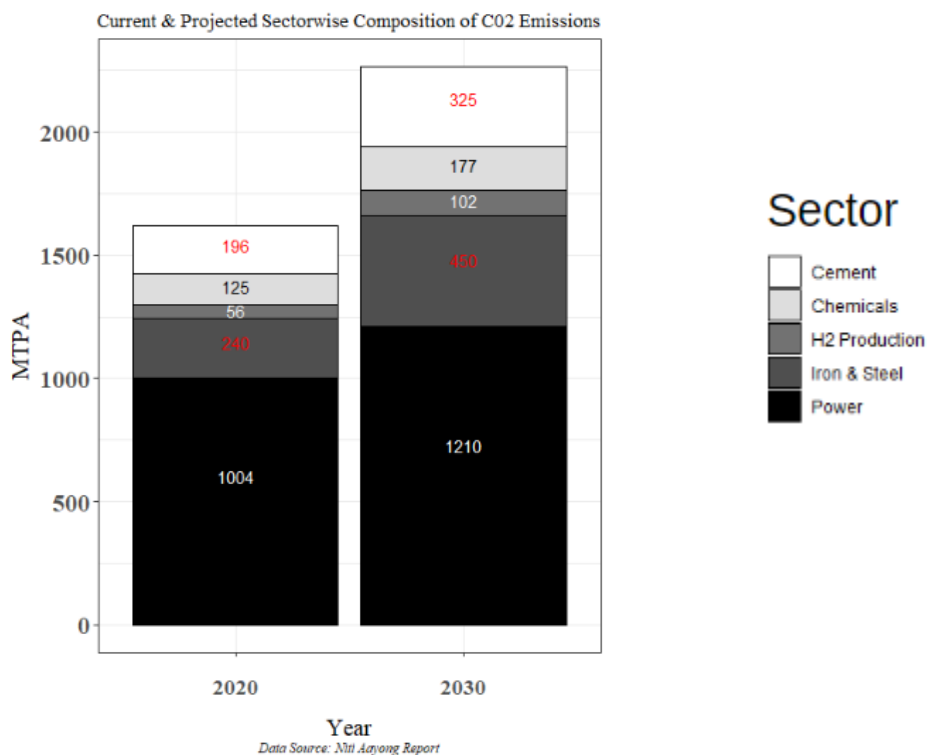


Figure 3: Sector wise split of industrial emission in India in 2020 and project values for 2030. Numbers in red highlight major emitting industries which will require CCUS to decarbonize
Source: Niti Aayog Report on CCUS

Industrial Emissions

Industrial emissions, along with emissions from the power generation industry, automobile and transportation, form a significant component of carbon emissions. Based on figures from IEA (International Energy Association), Industrial emissions account for nearly 23.5% of carbon emissions; Power generation accounts for 42.2%, the automobile and transportation sector account for 16.2% and agricultural and emission due to land use change account for 5.7% of the total carbon emissions. As we had discussed in the previous section, these emissions have been increasing rapidly, leading to a detrimental impact on biodiversity and society. These industrial emissions can be released in the form of – carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), sulfur dioxide (SO₂), nitrogen oxides (NO_x), particulate matter (PM), and volatile organic compounds (VOCs). A transition from burning fossil fuels to clean sources of energy will not suffice; in order to meet net zero commitment it is important to explore technological alternatives to address the problem of industrial emissions.

Fertilizers, iron and steel, cement, oil & gas, petrochemicals & chemicals, and fertilizers are the major contributors to industrial emissions. These industries are crucial for the development and

growth of any economy as they form the backbone of the economy. Renewable energy, in this case, is not the solution as it would only be to decarbonize these industries to a certain extent, ie, electricity needs; however, these industries emit a lot of pollutants as well into the atmosphere. Hence, these industries have been classified as “hard to decarbonize” industries and CCUS provides one of the only known technological alternatives to decarbonize such industries.

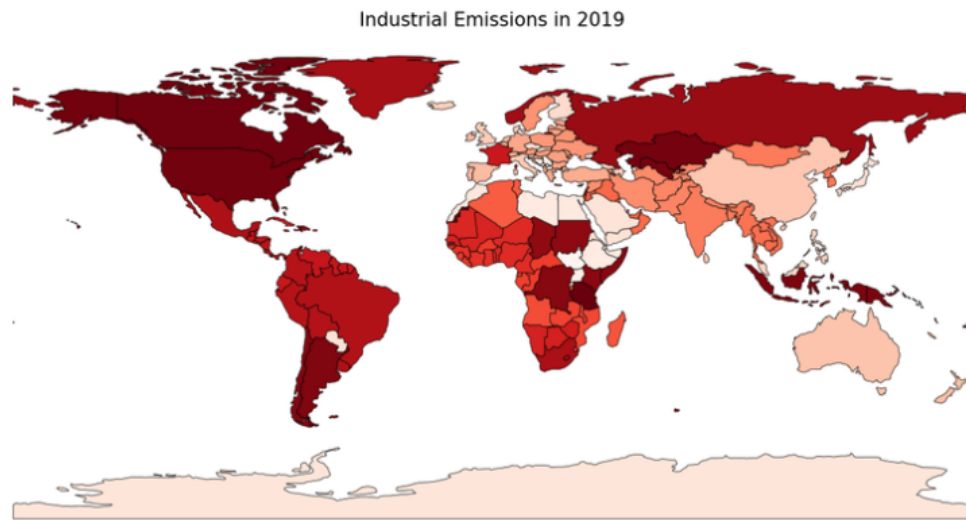


Figure 4: Choropleth map depicting industrial emissions across the world in 2019
Data Source: Climate Watch Data

We can see that developed nations have higher industrial emissions from the figure below; however, so does Africa. While we all know that the climate change crisis is a global issue and requires cooperative action; developed countries have been accused of shifting their responsibilities to the developing nations. The developed countries, as opposed to leading the way by doing, exploit and run away from environment protection. Of late, developed countries have found a "free to pollute" card by means of investing in projects and setting up factories in developing nations, thereby reducing emissions in the developed nation while adding the burden of reducing emissions outside. They have also been accused of "Carbon Leaking" – A practice wherein developed nations have set up factories in developing nations to manipulate the process of carbon counting; carbon is counted on a production level as opposed to where the goods/services are consumed; thereby, shifting manufacturing activities to countries with lower norms for environment protection – creating a detrimental impact on the environment.

Mapping Industrial Emissions in India

This study aims at studying whether CCUS is a viable technological alternative to decarbonize Indian industries from an economic, ecological and a policy perspective. An overview of industrial emissions and trend in emissions in India over past few decades is necessary before proceeding to understand the scope of CCUS in India. India is the seventh largest country by land area and the largest by population. India was also one of the fastest growing countries based GDP growth rates in the world recently. Structural change, i.e, a shift from an agrarian economy to an industrial or services oriented economy is accompanied with growth. Industries are emerging at a rapid rate to fulfill the demand for food and other goods. Large areas of land are being converted from forest and agricultural to commercial housing, spaces or factories. The environment is usually vastly neglected while making decisions. This sections aims to understand India from a comparative perspective vis-a-vis the world and run state level comparisons on find the trend in industrial emissions.

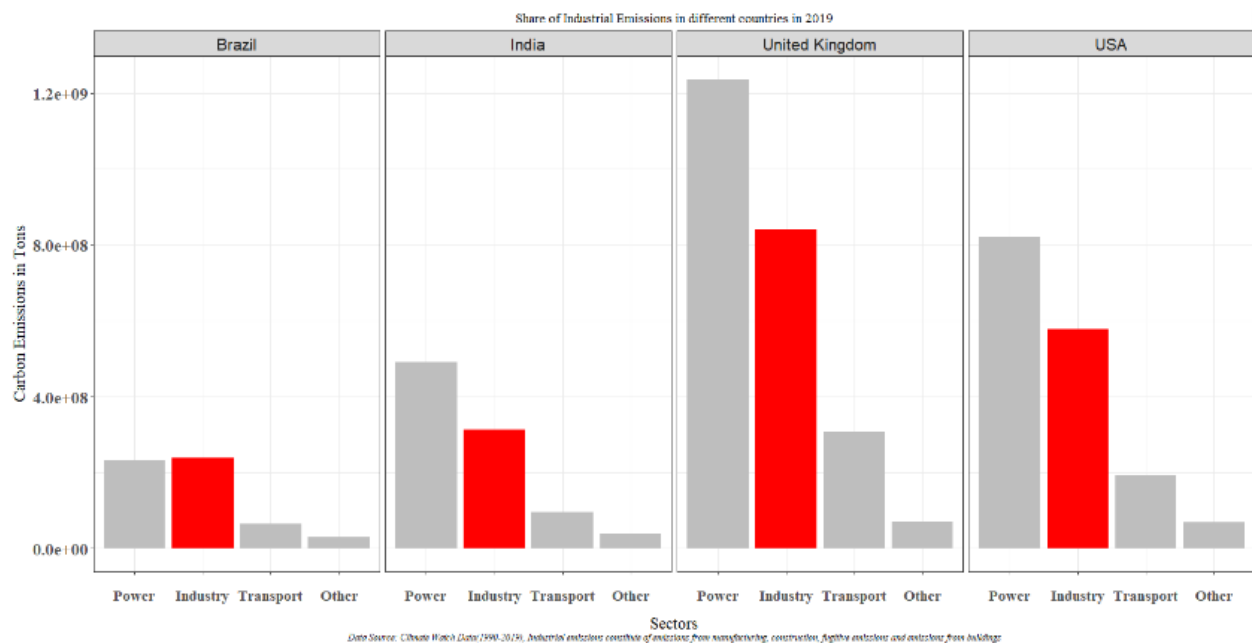
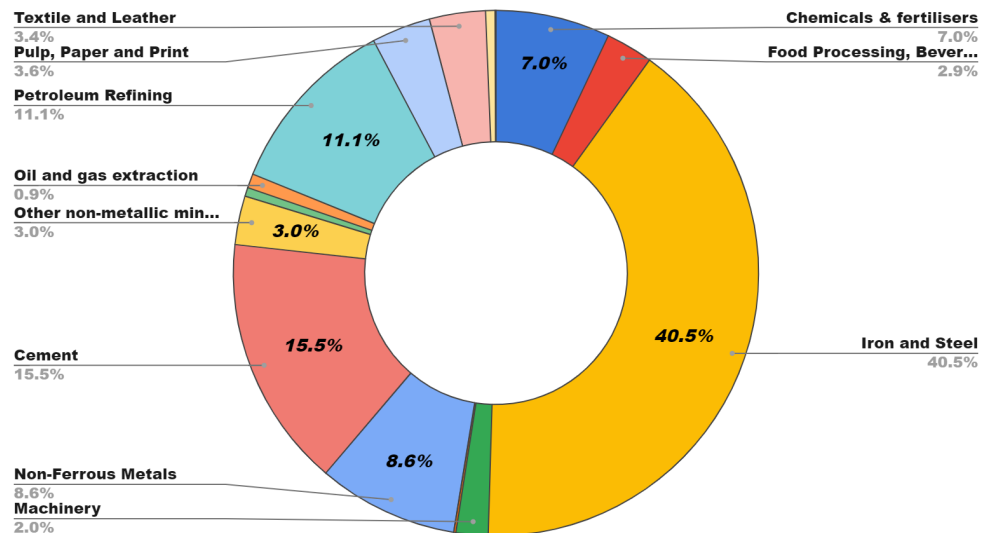


Figure 5: Comparing Industrial emissions in India with Brazil (Developing), UK and USA (Developed)

Data Source: Climate Watch Data

Despite having low per capita emissions India is the 3rd largest carbon emitter with 2654 million tons of CO₂ (China is the first with 10065 million tons, followed by USA with 5416 million tons). Cutting down on emission and enforcing strict environment protection laws is the need of the hour.

HIGHEST EMITTING INDUSTRIES IN 2005



HIGHEST EMITTING INDUSTRIES IN 2018

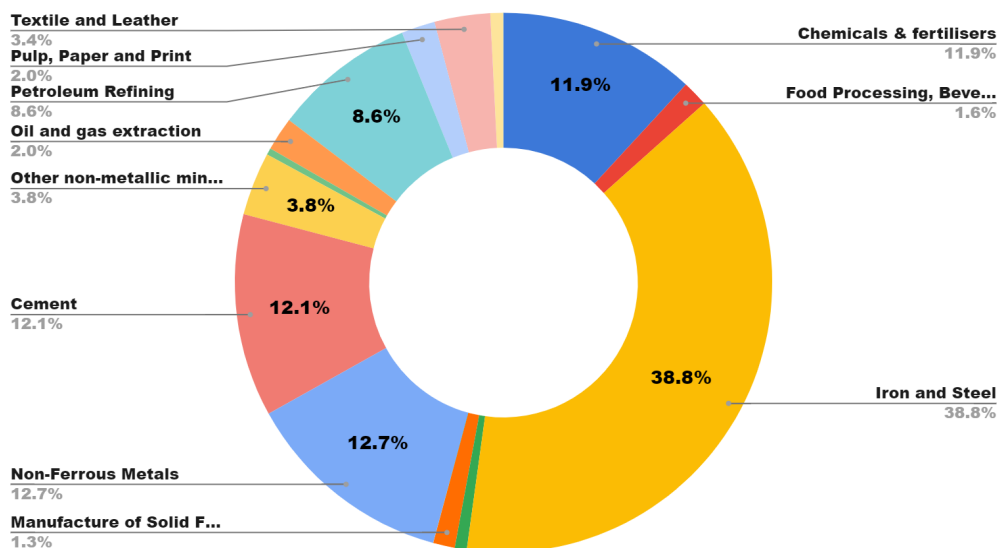


Figure 6: Split of Highest Emitting Industries in 2005 and 2018 in India
Data Source: GHG Platform India

The share of emissions from the chemicals and fertilizers and the oil and gas extraction industry has increased; both of are majorly polluting industries. The share of industrial emissions in the Iron and steel, cement and petroleum refining has declined. No major shift in trends or preferences can be observed between 2005 and 2018. It is interesting to observe the components of industrial emissions in India.

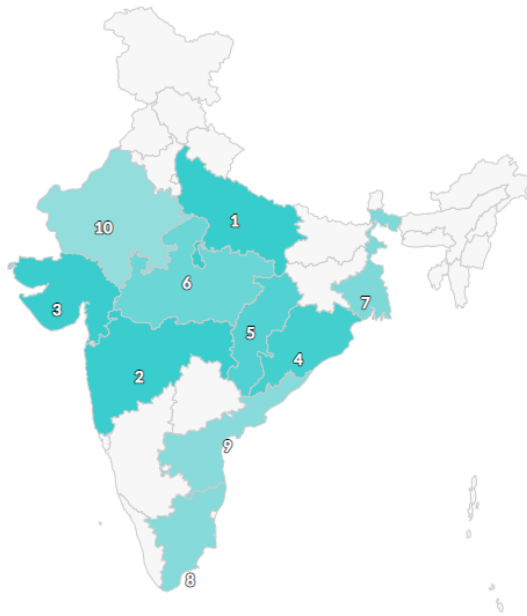


Figure 7: Choropleth Map with the top 10 carbon emitting states of India

Data Source: GHG Platform India

State/Union Territories (Top 4 and Bottom 4)	Average Growth Rate of Carbon Emissions from 2008 to 2018
Daman and Diu	3.23
Delhi	1.09
Nagaland	1.09
Assam	0.97
Madhya Pradesh	0.01
Tripura	-0.24
Arunachal Pradesh	-0.25
Chandigarh	-0.39

Table 1: Carbon Emission Growth Rates of the Top 4 Fastest and Slowest Growing Indian States
Data Source: GHG Platform India

Uttar Pradesh, Maharashtra and Gujarat are the states with the highest carbon emissions in India. The growth rate of carbon emissions or pollution, however, has slowed down in recent times in the states which previously had very high carbon emission, for example: Madhya Pradesh. Despite having extremely high levels of pollution, Delhi continues to witness high growth in carbon emissions. The southern states of Tamil Nadu, Kerala and Andhra Pradesh have performed in terms of limiting their carbon emission; they have one of the lowest average growth rates of pollution amongst other Indian states.

$$\left(\frac{\log \frac{Emitn}{Emit0}}{T}\right) = \beta_0 + \beta_1 \log(Y_{Emit0}) + \lambda_i + R_j + \epsilon_{ij}$$

=====	
Dependent variable:	

Growth_Rate	

Carbon_Emission	-0.00000 (0.00000)
Constant	0.436*** (0.138)

Observations	33
R2	0.078
Adjusted R2	0.049
Residual Std. Error	0.627 (df = 31)
F Statistic	2.641 (df = 1; 31)
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Note:	*p<0.1; **p<0.05; ***p<0.01

Table 2: Running an OLS regression on Growth rates of Carbon Emissions and its Initial Levels
Data Source: GHG Platform India

We ran a simple linear regression with the growth rate of carbon emissions as the dependent variable against the initial level of carbon emissions to test whether states which had high initial carbon emissions shown reductions in their carbon emissions, as well as to test for whether states which did not have high carbon emissions initially pollute at a faster rate. The numeric values as seen in Table 1 provide a better picture as no inferences can be drawn from the coefficient of interest here. The negative sign indicates to some extent that states which had lower emissions

either started polluting more or the states with higher emissions cut down, for the latter, Madhya Pradesh is a good example; however, we cannot make this claim since the results are not statistically significant.

Proposed Solutions to Rising Industrial Emissions

Improving Energy Efficiency

Increasing energy efficiency is an essential step towards reducing carbon emissions. Energy efficiency refers to using less energy per unit input for generating the same amount of electricity. Using energy more efficiently reduces greenhouse gas emissions while meeting the growing demand of energy. Renewable fuels are considered more efficient than fossil fuels; however, even renewable sources of energy can be made efficient through better absorption technologies in the case of solar energy and efficient turbine designing in the case of wind energy. Energy efficiency also comes with cost savings as opposed to large lumpsum investment for the creation of renewable energy projects, which makes this ideal for firms and industries which are run on the mechanism of cost savings.

Sustainable Agriculture

The practice of sustainable agriculture refers to the production of food without causing harm to the environment. It includes avoiding overuse of chemical fertilizers and pesticides and replacing them with natural fertilizers such as sewage sludge biochar (SSB). The SSB can help improve the soil fertility, the health of the plant and increase yields; and, since it is not using chemical fertilizers it is also helping in reducing carbon emissions. Sustainable agriculture also includes conservation of trees, forests and plants which act like carbon sinks.

Hydrogen Production

Hydrogen is an energy carrier which is available in abundance. It has the potential to take the place of fossil fuels and have carbon free methods of electricity generation. At present, there are various methods which could be adopted to produce hydrogen; some of these include steam methane reforming (not carbon neutral), while others include electrolysis, which is a carbon neutral method. Hydrogen has a lot of potential, particularly in the electric vehicles segments where hydrogen is proposed to be used as a fuel to power the engine; this will eliminate the emissions which are generated by the tailpipe of the car and reduce carbon emissions.

Renewable Energy

Renewable energy refers to energy generated from natural sources which can be replenished at a rate faster than they are consumed. Burning of fossil fuels is a major contributor to increasing carbon emissions; with a switch to renewable energy technologies, we can reduce these carbon emissions. Starting out as costly solutions to an urban living, renewable energy, over time has become cost effective making it more attractive to firms and industries which are willing to reduce their carbon footprints. However, in the case of renewable energy, a transition from fossil fuels to renewable energy may not be enough; this is because there are some processes of industries which cannot be electrified or cannot rely solely on renewable energy due to their scale. Even otherwise, the by-product of the production process, which is usually a bunch of pollution emitted into the atmosphere, will still exist; CCUS or carbon capture utilization and storage is the technological solution to.

Introduction to CCUS : Carbon Capture, Storage, and Utilization

The Industrial Ecology Framework

We have used an industrial ecology framework for this paper. Production and consumption is an integral part of all production processes, what is unique about this framework is the incorporation of by-products or waste materials in the production process cycle. The domain of industrial ecology looks at industrial processes and the industries operating in concert with the environment around them. The evolution of economy, environment, and technology has produced ideas and techniques that could be well incorporated into the modern industrial facilities to ensure sustainable management of environmental resources. A loop of processing raw materials, subsequent conversion of those raw materials and generation of waste which is utilized back into the production cycle through different channels is put in place through this framework. The raw material in the case for decarbonizing industries is carbon emissions—making this a carbon neutral production framework. One of the areas that this domain explores is that of decarbonizing since industrial facilities have historically been the major source of carbon emissions into the environment.

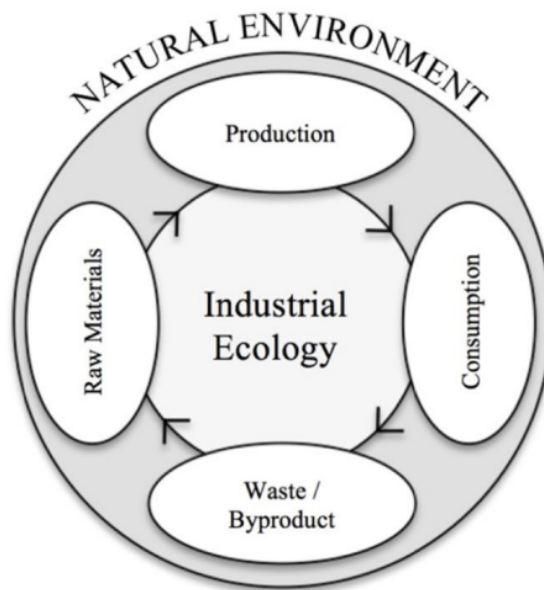


Figure 8: The Industrial Ecology Framework explained using a diagram
Image Source: Journal of Sustainable Development

About CCUS

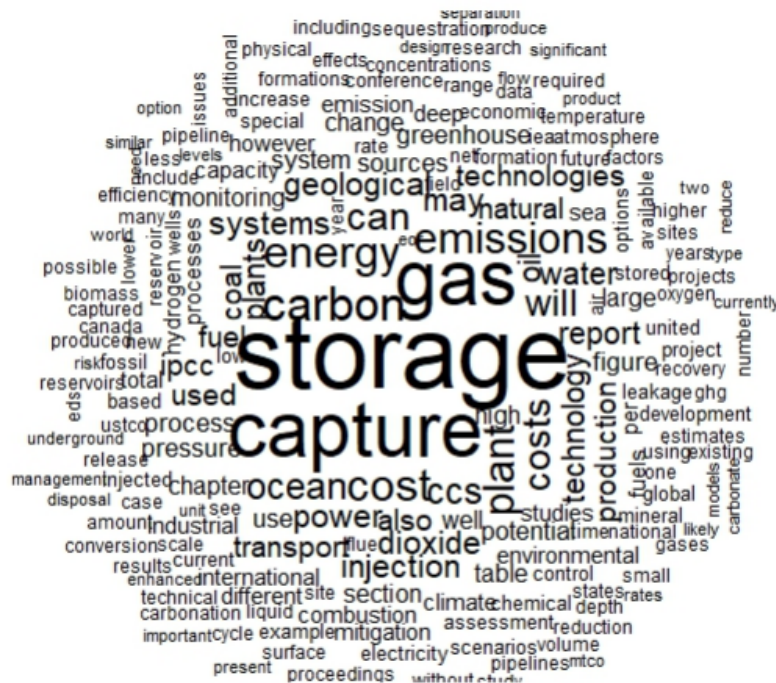


Figure 9 : A Wordcloud depicting the words which were used most frequently in the IPCC report titled 'Carbon Dioxide Capture and Storage'

A range of technologies fall under the broad umbrella of Carbon Capture Utilisation and Storage. It ranges from activities that capture the carbon before emitting it into the atmosphere or more recently capturing carbon dioxide from the air, formally called Direct Air Capture. Post this capturing, the carbon can potentially be utilized again as an energy input for the industry, much like the natural ecosystems. The scope of the technology is currently being discussed mostly in the context of fossil-based power generation but the evolution of the technology promises this solution to be applied in other sectors as well.

The CCUS or Carbon Capture Utilization and Storage will play a pivotal role in decarbonizing the industrial sector in the future. Fossil fuels have been since ages an integral part of the

production process in industries; not only are the fossil fuels used as a source of energy but they have also been used within the process itself. This makes the industrial sector hard to electrify – CCUS provides a technological solution to this problem. The role of CCUS magnifies even further due to the incapability of renewable energy to achieve net zero industrial emissions.

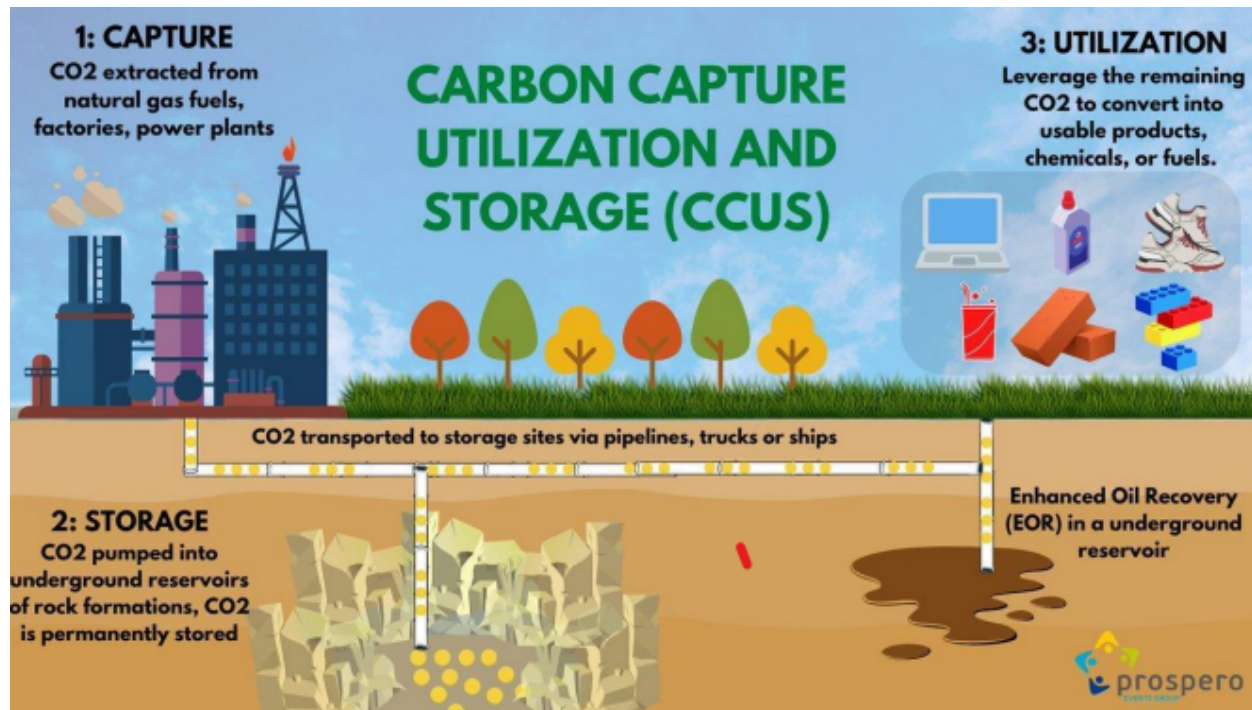


Figure 10 : The Process of Carbon Capture, Utilization and Storage

Image Source: Nextias

The Emergence of CCUS

CCUS projects and technology are still at an infant stage of development. While the concept of CCUS was first proposed back in the 1970s by JW Smith. In The 1980s the first ever commercial CCUS project was sanctioned by the government of Canada. This Canadian project (Weyburn-Midale CO₂ Project) became the first project to use CO₂ for enhanced oil recovery. The first ever commercial CCUS project was the Sleipner project in Norway which began injecting CO₂ into an underground reservoir back in 1996. All of the early 21st century countries built their first ever running CCUS projects. These ranged from the Salah project in Algeria, the Gorgon project in Australia to the Petra Nova project in United States of America which became the world's largest post-combustion carbon capture facility in 2005. Fast forward 10 years, in

2015, The Paris Agreement on climate change recognizes the importance of CCUS in achieving global emissions reduction targets; CCUS was now deemed as an essential technology to battle climate change and reduce carbon emissions. Post the agreement, many European countries announced research programs and funding in the fields of CCUS for innovation and development of such projects. In 2020, the Norwegian government plans to build a large scale CCUS cement plant; the first of its kind. Finally, Dalmia Bharat Cement Limited, an Indian cement company has planned to open up a cement CCUS plant in Tamilnadu, India.

Carbon Capturing : Process and Technologies

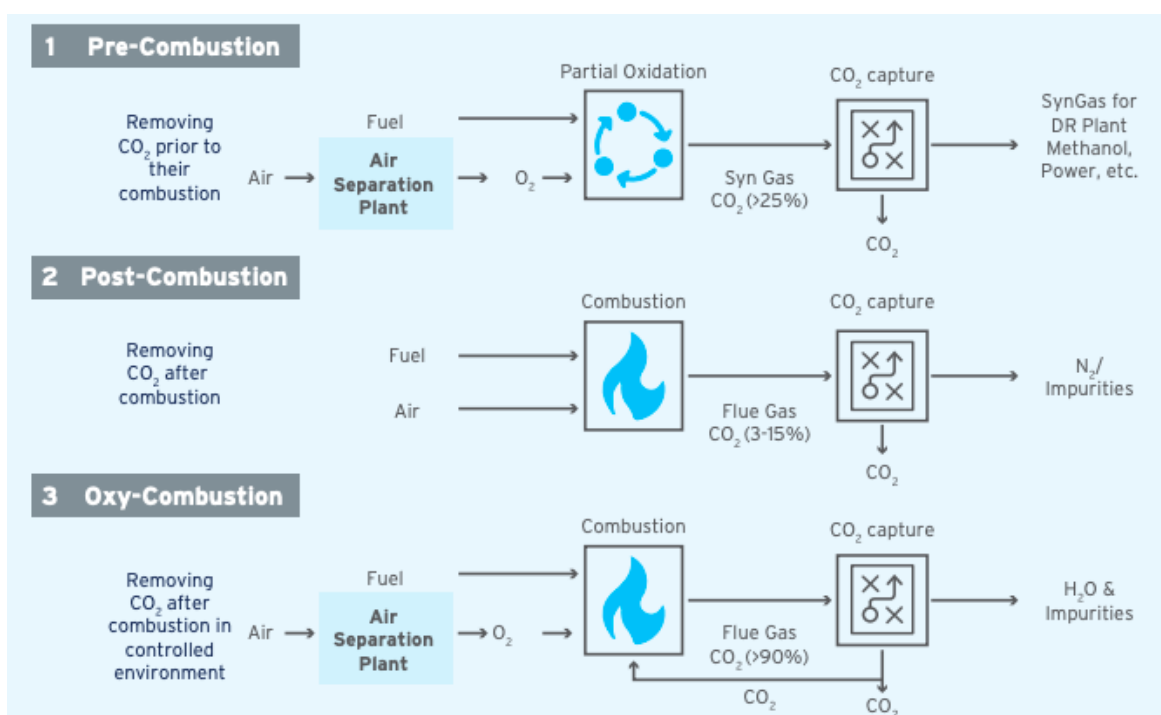


Image 11 – Source:- Niti Aayog Report (November, 2022)

Pre-Combustion Capture

In the case of Pre-Combustion Carbon Capture Technologies, CO₂ is removed from the fossil fuels before it is combusted. This method of CO₂ capture involves the oxidation of the fossil fuel in air/oxygen and steam. In this method, Syngas (Synthesis Gas) is also produced. The Synthetic Gas (Syngas) is a mixture of Water (H₂O), Carbon Monoxide (CO), Carbon DiOxide (CO₂) and smaller components of other gases. The syngas then undergoes the water-gas shift reaction that

converts CO and water (H₂O) to Hydrogen Gas (H₂) and CO₂. The product CO₂ can be captured, transported, sequestered or utilized. Post the removal of CO₂, the H₂ rich syngas can be deployed in multiple clean energy system applications.

Post-Combustion Capture

The Post-Combustion Carbon Capture technology involves the capturing of CO₂ from the flue gas streams after its combustion at low pressure. The flue gas which mostly comprises CO₂, Nitrogen, and some other oxygenated compounds, which is treated with an amine solution and the CO₂ gas is stripped out, compressed and cooled in liquid form. This method has been developed and deployed commercially for many years in chemicals and refining processes. This method can be used to capture CO₂ from gas fired power plants and other such large stationary sources of CO₂, for instance, smelters, cement kilns and steelworks, etc.

Oxy-Fuel Combustion

In this process, the hydrocarbon (fossil fuel) is combusted in a nearly pure oxygen environment instead of air. This is a relatively new carbon capturing technology where CO₂ rich flue gas is used to dilute oxygen which also results in lower NO_x emissions.

Direct Air Capture (DAC)

The Direct Air Capture (DAC) Technologies extract CO₂ directly from the atmosphere. An engineered mechanical process is used to build a CO₂ collector. The CO₂ loaded air is drawn in through the fan inside the collector. The CO₂ particles are trapped using a filter and the CO₂ is collected in a separate container at a temperature of 100 degree celsius. Currently, there are around 18 direct air capture plants operating worldwide which almost capture 0.01 Mt of CO₂ per year.

CO2 Storage in Geological Formations

Enhanced Oil Recovery (EOR)

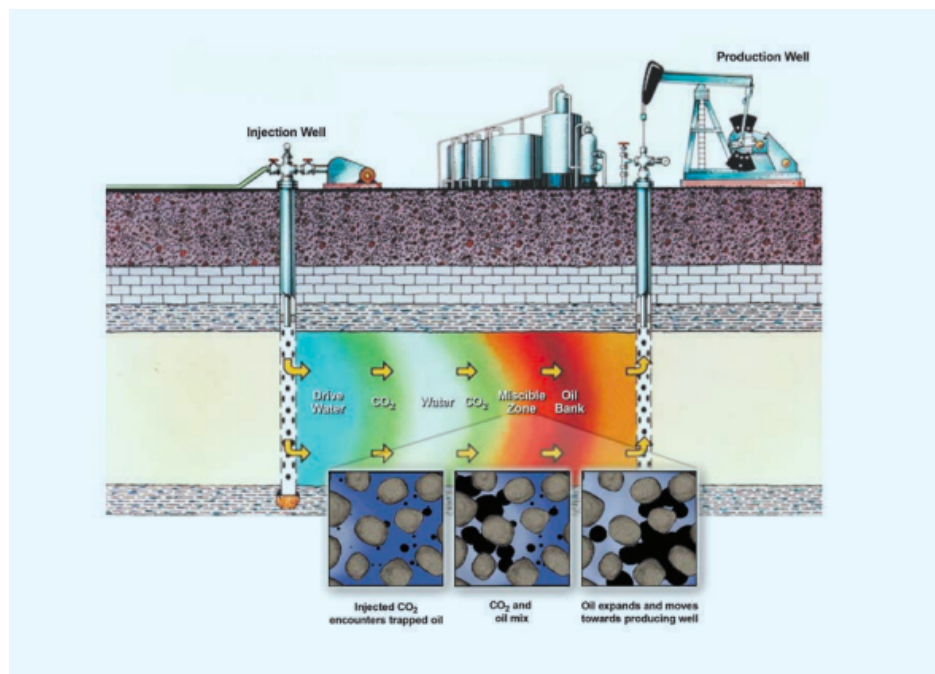


Image 12 – Source:- Enhanced Oil Recovery, NITI Aayog Report, 2022

Enhanced Oil Recovery refers to the extraction of oil from oil fields that is impossible to extract otherwise. The oil recovery techniques are categorized into Primary, Tertiary, and Secondary techniques. EOR is a tertiary oil recovery technique used to further improve the production performance of wells. In CO2 EOR, compressed CO2 is injected into the oil reservoir which swells the oil and drives it away from the rock formations towards the production wells. This technology has been studied and applied globally as well by Indian oil exploration companies.

Enhanced Coal Bed Methane Recovery (ECBMR)

The process of Enhanced Coal Bed Methane Recovery (ECBMR) involves the injection of CO2 into unmineable coal seams under supercritical conditions. This pushes the Coal Bed methane towards the production wells and the CO2 is stored geologically for potentially permanently. The Niti Aayog Report suggests that the location of coal fired power plants near coalfields makes it a viable business opportunity since the recovered methane can help offset the cost of carbon capture.

Deep Saline Aquifers

Deep Saline Aquifers possess the potential to store large quantities of CO₂ as they are spread over large areas. They do not have a direct economic benefit as in the case of EOR and ECBMR but feasibility studies suggest this as an alternative storage option. They consist of porous rock formations which contain large amounts of unusable salt based water. When injected, CO₂ gets trapped into the pores due to its lower density as compared to brine water. In the case of India, large amounts of deep saline aquifers have been identified to geologically store CO₂.

CO₂ Utilization

The capture CO₂ can be utilized in various sectors such as:

- Green Urea:- The total urea consumption in India is more than 30 metric tonnes per year. It can be producing CO₂ and H₂, and can replace the conventional natural gas based production of urea and also help abate the import bill of the sector.
- Food and Beverages (F&B) :- The captured CO₂ can also be utilized in the F&B sector for its application in the production of carbonated beverages, dry ice, atmosphere packaging, etc.
- Chemicals :- The use of CO₂ for the production of methanol and ethanol has existed for many years. Both these chemicals have large fuel-substitution possibilities and are thus crucial for the manufacturing industry at large. It can also help reduce the import bill as in the case with green urea as the scope to utilize CO₂ using this pathway is large-scale.

There are various other sectors such as Building Materials, Polymers, Bioplastics, Enhanced Oil Recovery, etc., where the use of CO₂ as an input has been well demonstrated across the world.

Carbon Transportation

A safe and reliable transportation of CO₂ from the capture site to the storage/utilization site is a necessary component of the CCUS process. Tanks, pipelines, ships, and other such large transportation mechanisms are used to safely transport the captured carbon. Among all of them, pipelines are the most used due to their scalability to transport CO₂ much like any other gas. Ship transportation is being considered as an alternative offshore transportation, with small-scale CO₂ ship transportation already taking place in Europe. Large Scale transportation of CO₂ is going to be similar to the transportation of LPG, which has experienced a massive scale supply and transportation in the past 70 years. CO₂ pipelines need a strong risk assessment before CO₂ transportation takes off at a large scale as the associated environmental risks are extremely large and hazardous.

A global picture of CCUS

A growing trend can be seen in the number of additional CCUS sites built year on year. There are nearly 200 projects in the CSS facilities pipeline; which is achieved through a phenomenal 44% growth in the number of CCS projects in 2021. There are plenty of notable CCUS projects that have been proposed with enormous capturing technologies in recent years by countries in Asia and Europe. At present, a majority of the developed and running projects are clustered around the United States with others scattered across the globe. India has also started work on CCUS projects of late; however, a lot of planning and feasibility studies, similar to that of the NITI Aayog report, must be conducted in this field. The CCUS facility numbers and their capture capability have been increasing at a record rate; Since 2017, capture capacity has grown at a compound rate of over 34 per cent per annum. There has been a near double of those projects in their advance development strategies in terms of their capturing capacities. The top countries based on facility count are given below.

Country	Number of Projects since 2021
USA	34
Canada	19
UK	13
Norway	8
Australia	6
Netherlands	6
Iceland	6

Table 3 :Countries with the Highest Number of CCUS Projects Since 2021

Image Source: Global CSS Institute

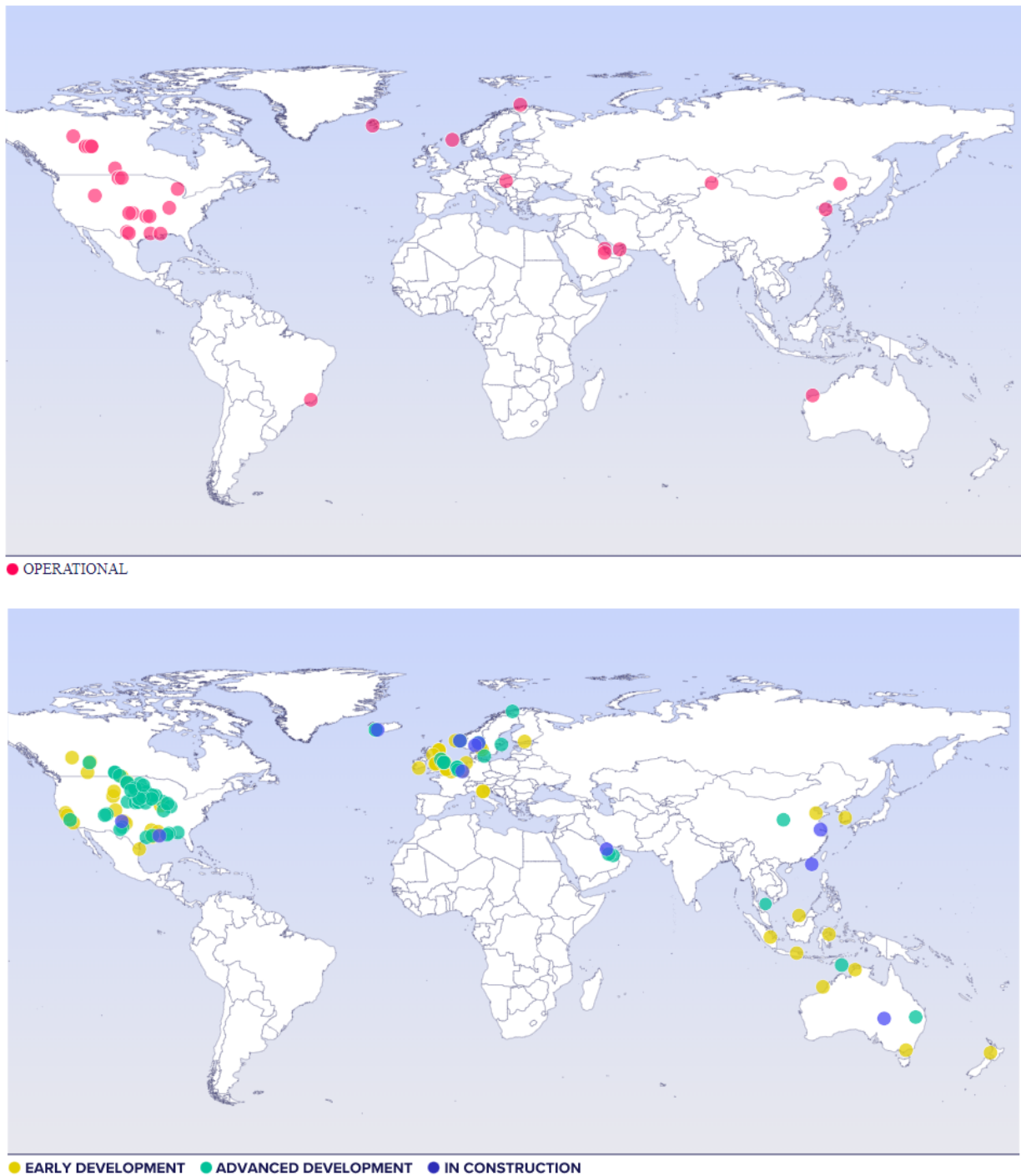


Figure 13 : Number of Operational CCUS facilities and Number of Facilities in Development
Image Source: Global CSS Institute

CCUS in India : Evolution, Risks and Benefits, Policy Framework

Evolution of CCUS discourse in Indian policies

The Indian climate and energy policy has historically focused on energy efficiency and renewable energy as the major contributor to mitigate climate change. Despite lacking a critical thrust in terms of policy framework, India has managed to enable some institutions to foster research and development of Carbon Capture, Utilization, and Storage (CCUS). These efforts have been funded by both the government and the industry.

The setting up of **Indian CO₂ Sequestration Applied Research (ICOSAR) Network** has been considered as a foundational step for institutional research and development in the areas of Carbon Capture and Storage in India. This network was established in 2007 under the Department of Science and Technology, Government of India, with an aim to coordinate the activities in research and development of the technology.

The **National Action Plan on Climate Change, 2008** highlighted concerns relating to the applicability of CCUS in India. These concerns included the high cost structures and (in)stability of CO₂ storage repositories.

Later in 2012, India presented a demonstration project in its **Second National Communication (SNC)** to the United Nations Framework Convention on Climate Change (UNFCCC).

In 2015, the COP 21 took place in Paris where **Mission Innovation** was announced and Carbon Capture was considered as Innovation Challenge 3 and India recognized the action on IC3 through global R&D coordination. It became a part of a 24 country consortium which agreed to finance these R&D activities.

India's Biennial Update Report (III) in 2021 included the vision and ambition to set up CCS facilities in carbon intensive industries. It also included the demonstration of CCS facilities under initiation by ONGC, BHEL, NALCO, etc.

The **NITI Aayog launched a report** titled “Carbon Capture, Utilization and Storage (CCUS) - Policy Framework and its Deployment Mechanism in India” in November, 2022.

Environmental Risks

Seismic Risks

(Zoback and Gorelick, 2012) argue that there is a high probability that earthquakes will be triggered by injection of large volumes of CO₂ into the brittle rocks commonly found in continental interiors. Because even small- to moderate-sized earthquakes threaten the seal integrity of CO₂ repositories. In this context, large-scale CCS is a risky, and likely unsuccessful, strategy for significantly reducing greenhouse gas emissions.

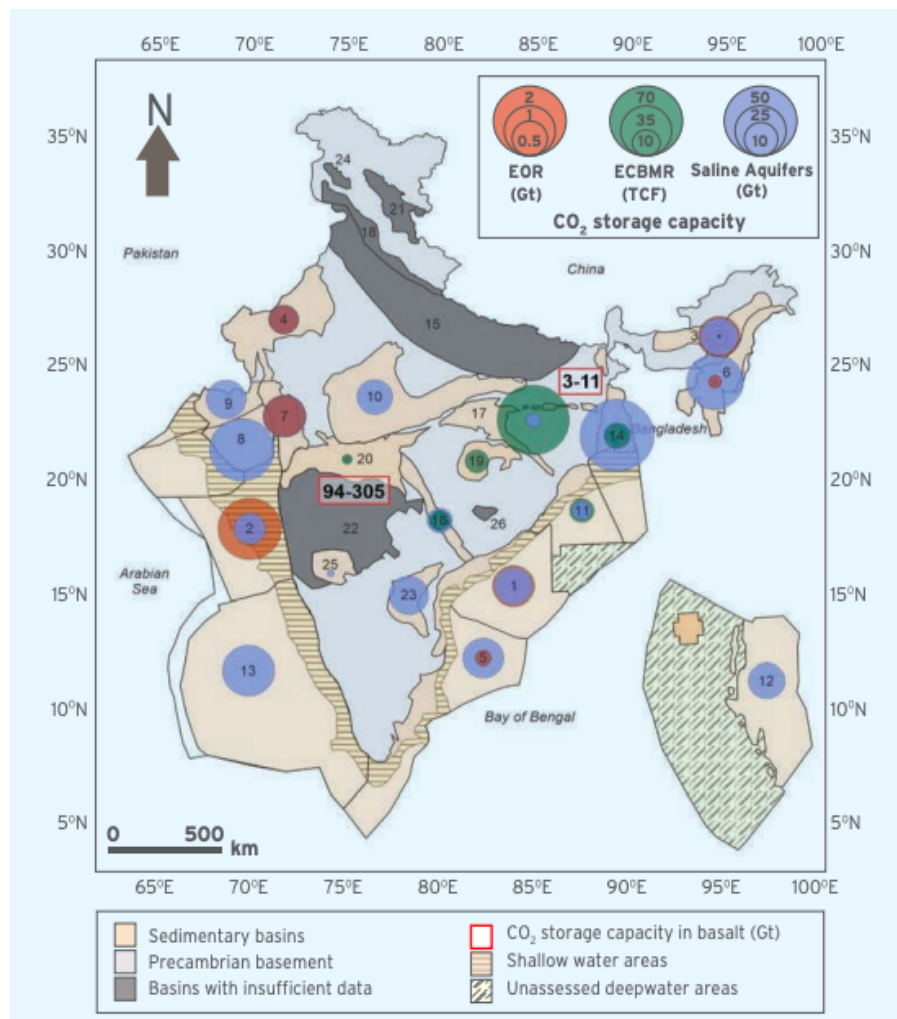


Figure 14:- Sedimentary Basins of India and CO₂ storage sites across India (Source : Niti Aayog)

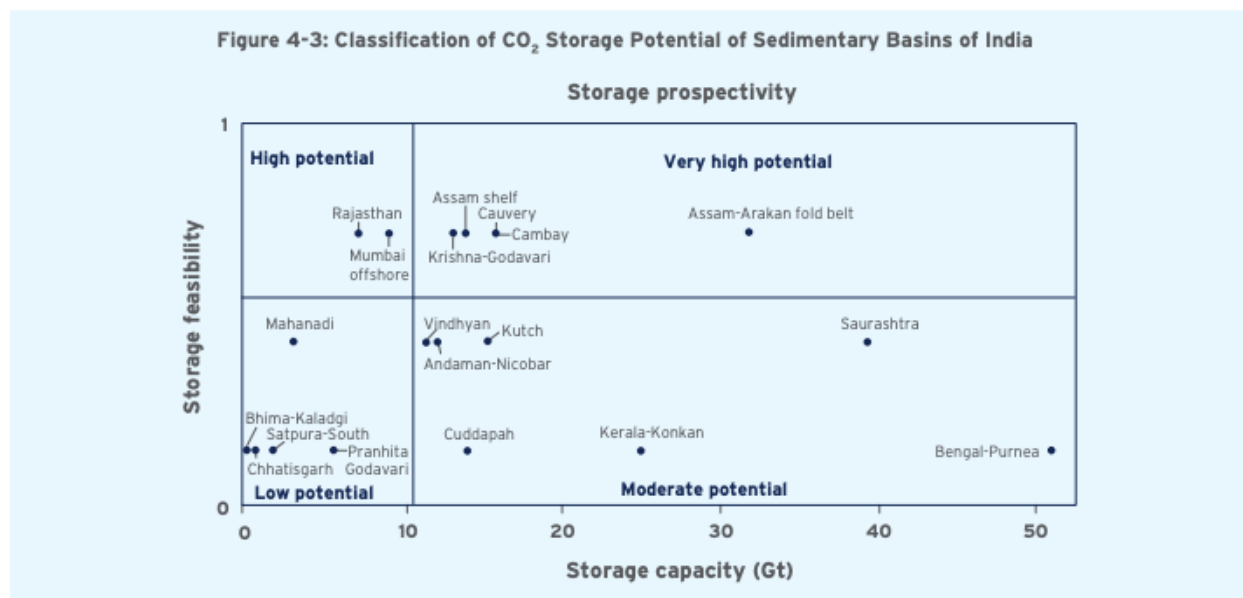


Figure 15:- CO₂ storage sites across India : Capacity and Feasibility (Source : Niti Aayog)

This map and the above graphic is a distribution of the CO₂ storage sites as assessed in the Niti Aayog Report, 2022. Most of the CO₂ storage avenues would be the Deep Saline Aquifers method in the sedimentary basins of India. As per the study by (Zoback and Gorelick, 2012), the injection of large volumes of CO₂ creates a potential seismic risk, which in turn threatens the seal of CO₂ reservoirs underground. In this report, we try to plot the major CO₂ storage locations (as per the NITI Aayog Report, 2022) in India on the Seismic Map of India (see next page) to highlight the potential seismic concerns with large-scale CO₂ storage. Among all the major CO₂ storage sites, the site at Assam-Arakan fold belt (32.3 Gt) falls under Zone V on the Seismic Map of India and thus is already an earthquake prone area.

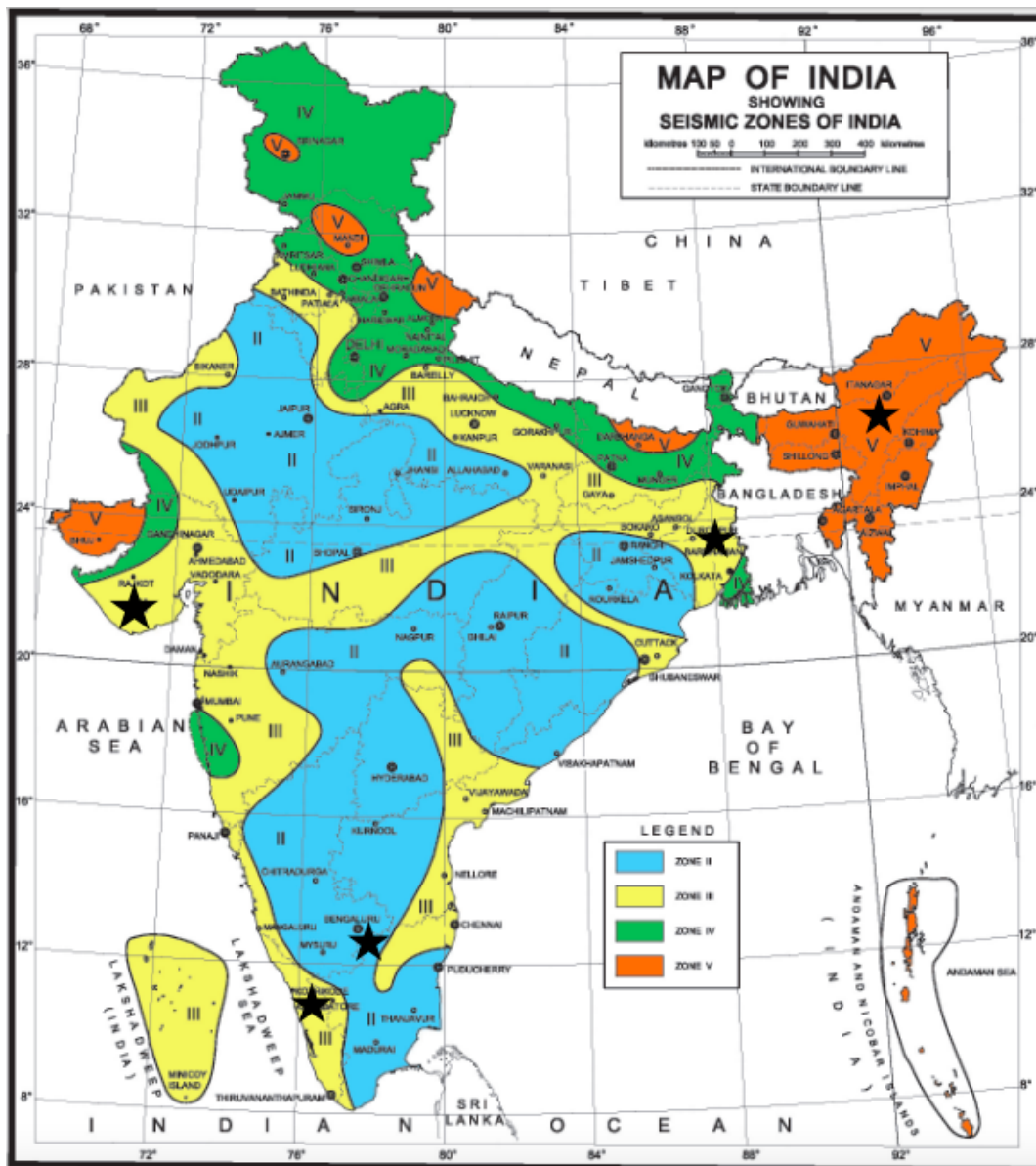


Figure 16:- Seismic Map of India and the major CO₂ storage sites

Potential Groundwater Shortages

The additional use of water needed for CO₂-EOR operations can have a potential negative impact on water resources in regions with scarcity of freshwater. Enhanced Oil Recovery (EOR) processes use water-soluble chemicals that may be toxic to organisms and carcinogenic to humans if they are found in sufficient quantities in groundwaters.

Social Risks of CCUS in India

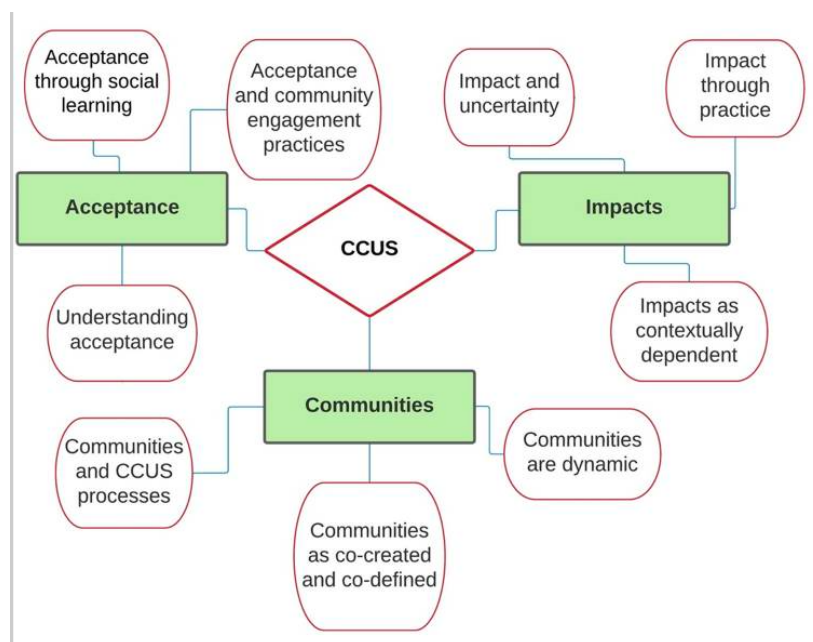


Image 17 – Source:- Nielsen.et.al. 2022

Nielsen.et.al. 2022 presented a systematic meta-narrative review of published studies on community acceptance and social impacts of Carbon Capture, Utilization and Storage (CCUS) projects across the world. All these projects had engagements with communities and the authors attempted to conceptualize the community acceptance of these projects. The other area the authors explored were the communities itself, if and how these communities were represented as stakeholders, etc. The third scope related to the contestation of these projects on grounds of high uncertainty and feasibility.

Such a study becomes very much relevant in India where mega projects continue to displace communities with steady rationalization from the state and its arms. Social justice implications of these mega projects are large and undemocratic as the communities whose livelihoods and local ecology are used for extractive purposes, and are left with no other secure livelihood. In the case of CO₂ EOR and ECBMR storage, the idea of resource metabolism and extraction becomes much more direct as the economic benefits are direct and there exist institutions to exploit these opportunities. Along with this, the added layer of seismic risks can leave households and communities with higher uncertainty and contestations regarding such projects.

Case Studies

Dalmia Cement (Bharat) Limited - Dalmiapuram, Tamil Nadu

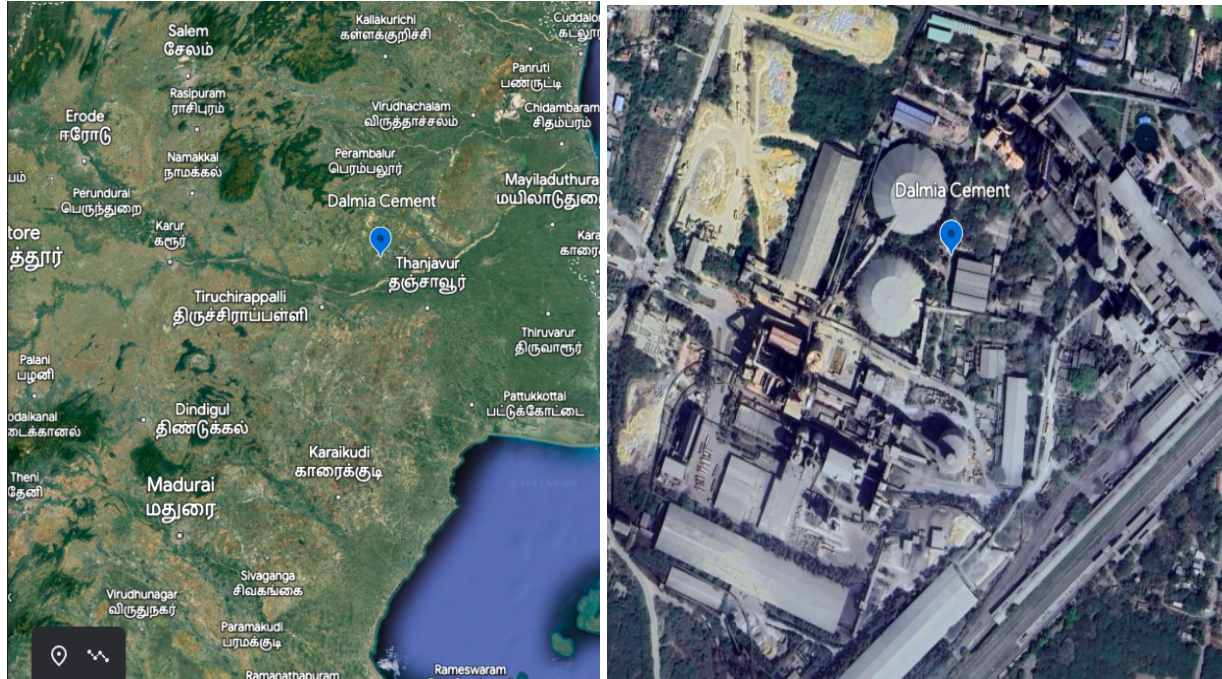


Figure 18 :- Satellite View and Location of the Dalmia Bharat Cement Factory, Dalmiapuram, TN

Cement is considered as one of the most carbon intensive and hard to abate industrial sectors of the economy. Dalmia (Bharat) Cement Ltd., the fourth largest Indian cement manufacturer which is steering its operations towards carbon neutral production of cement in the country. Currently, its emission rate measured by kilograms of CO₂ emitted per tonne of cement produced is around 546 kg/tonne. This is well below both the national average (650kg/tonne) and the global average (600 kg/tonne).

In 2019, the company announced that it will build a 500,000 tonnes of carbon capture facility at its cement manufacturing plant in Tamil Nadu. Such a large-scale CO₂ capture facility in a cement plant is unprecedented globally. The company also plans to be carbon negative by 2040 due to its various sustainability and conservation efforts.

Koyali Refinery - Vadodara, Gujarat

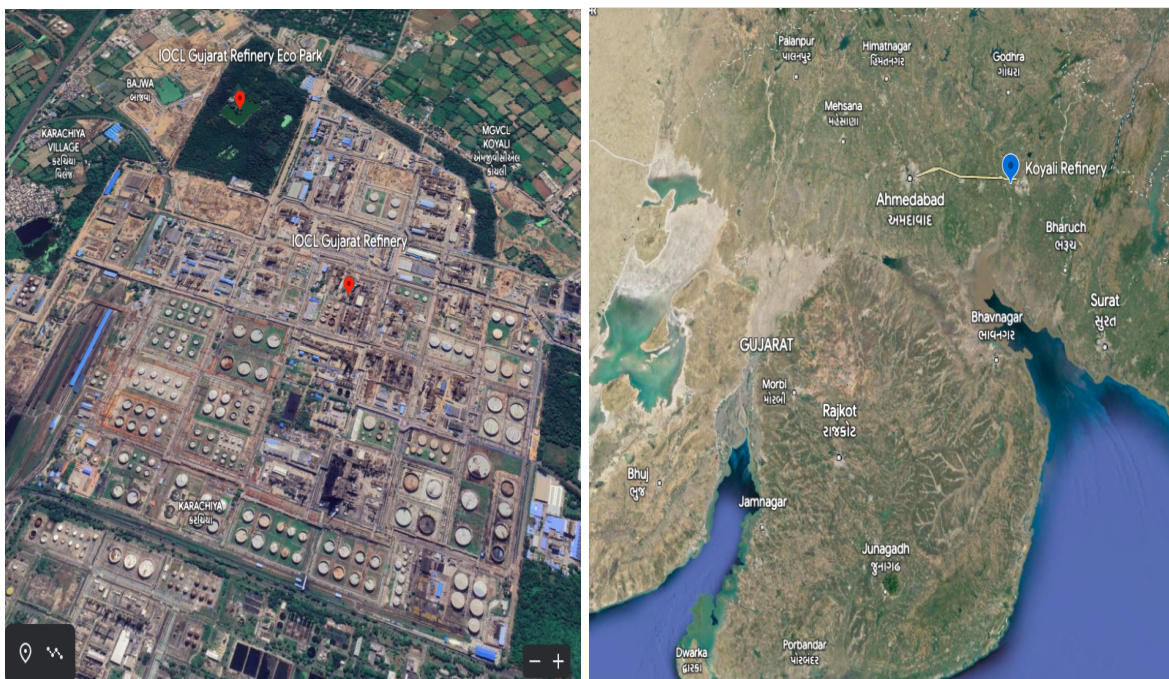


Figure 19:- Satellite View and Location of the Koyali Refinery, Vadodara, Gujarat

The Koyali Refinery (Gujarat) is an oil refinery owned by the Indian Oil Corporation Ltd. It is located at Koyali in the Vadodra District, Gujarat. It functions as a 13.7 million tonne integrated refinery-cum-petrochemical complex. Dastur Energy, a global leader in CCUS activities recently ran a technical and economic feasibility check for an in-house CCUS project to decarbonise the refinery operation. The CCUS project aims to capture almost 700,000 metric tonnes per annum of carbon dioxide (CO₂) from its Steam Methane Reforming (SMR) based Hydrogen Generation Units. This will produce clean hydrogen which is one of the major alternative clean fuels in consideration.

The project has been funded by a grant from the United States Trade and Development Agency (USTDA). The feasibility report suggests that the captured CO₂ would be utilized for ONGC's Gandhar oil fields for Enhanced Oil Recovery operations. A portion of the captured CO₂ would also be utilized in the food and beverage sector.

Results and Discussion

The project findings suggest that CCUS can be a potential solution to decarbonise the Indian industries, though nowhere in the near future. The existing technologies to do so are extremely costly and unavailable in the country. Cooperated global R&D efforts and investments can foster this large-scale solution in a country like India.

Also we find that like other such projects, there is a need to assess the likely environmental, safety, and health impacts in case of a CO₂ leakage event. The evaluation of the required remedial actions and their feasibility is thus necessary. We find that social and ecological impacts of the projects also need to be considered while choosing the storage sites as community acceptance and awareness is crucial.

Despite efforts by climate change economists like William Nordhaus (among many others) to model the cost of carbon, there is no price of carbon due to lack of carbon pricing or credit markets in India for the same. The economic costs of setting up carbon capture projects is very high with the economic benefits unknown. This makes it hard for a cost-benefit analysis to evaluate the economic feasibility of most projects. Since, CO₂- EOR projects have developed capturing and injecting technologies for many years, those are affordable than other types of projects.

The NITI Aayog report released in 2022 also suggests that carbon markets would incentivise CCUS projects and recommends establishing a Carbon Capture Finance Corporation (CCFC) to support CCUS projects through their lifecycle.

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